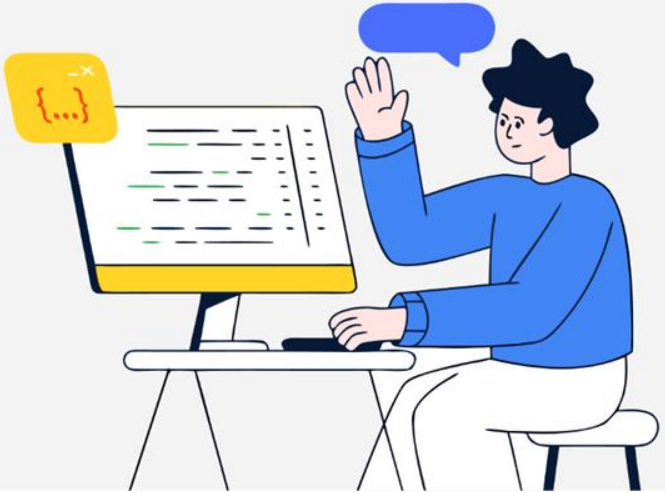
Build with AI

Solution Challenge



Team Details

a. Team name: TechMedics
b. Team leader name:Vinayaka G C
c. Problem Statement: [Rapid Crisis Response] Accelerated Emergency Response and Crisis Coordination in Hospitality

Brief about your solution

The Problem

- Black Box” Anxiety: Callers experience panic due to no visibility of ambulance location or arrival time.
- Connectivity Barrier: Emergency solutions fail in remote areas and highways with poor internet access.
- Dispatch Delays: Manual coordination and verbal address errors delay critical response during the Golden Hour.

Our Solution – Tech-Medics 108+

- Real-Time Visibility: Live ambulance tracking provides clarity and reassurance to victims.
- Offline Reliability: Location information is transmitted even without internet connectivity.
- Smart Dispatching: Automated nearest-ambulance assignment significantly reduces response time.

Opportunities,Differentiation & Viability

Differentiation

- Live ambulance visibility replaces blind waiting
- Automated dispatch replaces manual coordination
- Works even without internet connectivity
- Calm, emergency-focused user experience

Opportunities

- Deployable for government 108 & Smart Cities
- Valuable for insurers via faster response outcomes
- Scalable to private hospital ambulance fleets

Viability

- Exact GPS eliminates location ambiguity
- Transparency reduces panic and repeat calls
- Automation cuts response time from minutes to seconds

List of features offered by the solution

Live Ambulance Tracking
- Patients can see where the ambulance is and when it will arrive.

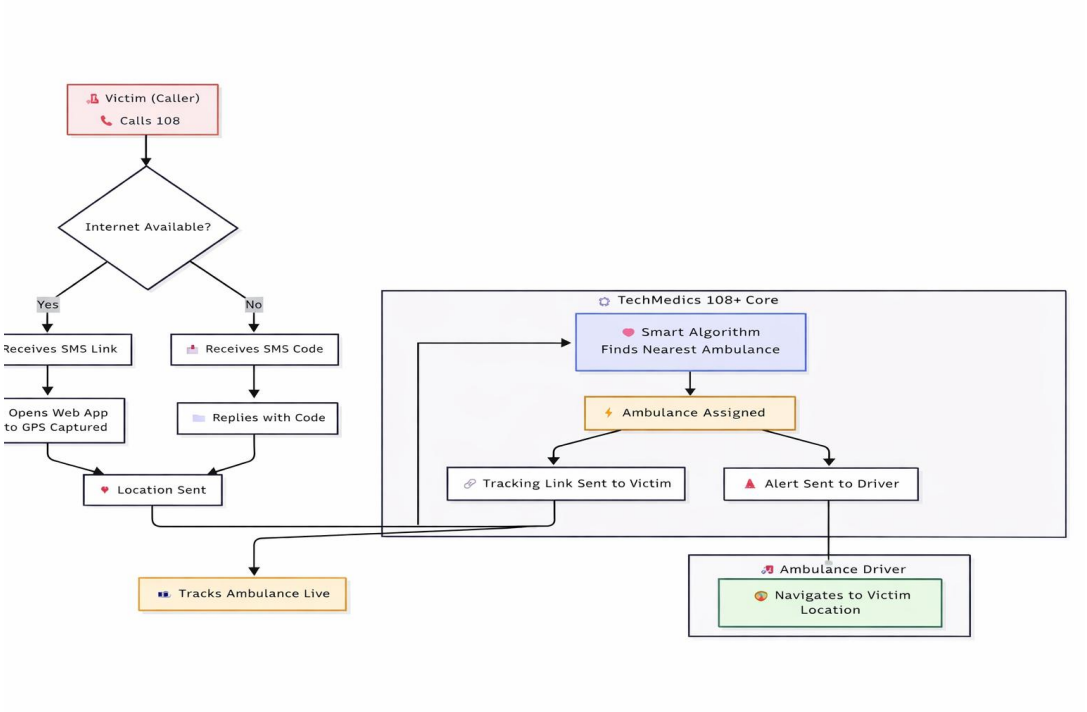
Works Without Internet
- Shares location using SMS when mobile data is unavailable.

Faster Ambulance Assignment
- Automatically sends the nearest ambulance to the emergency.

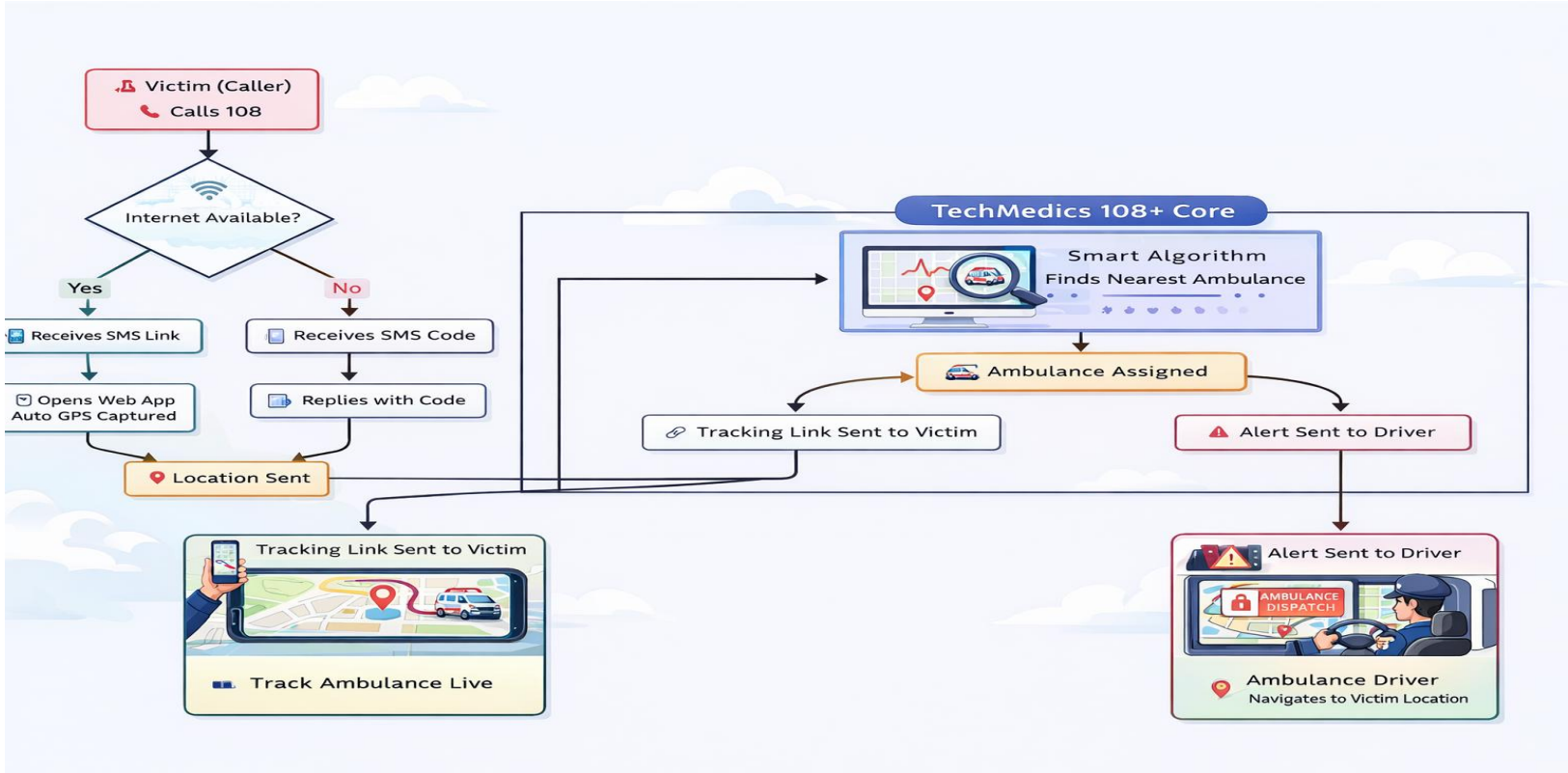
Driver-Friendly System
- Ambulance drivers get clear directions and status updates.

Simple & Calm Design
- Easy-to-use interface for stressful emergency situations.

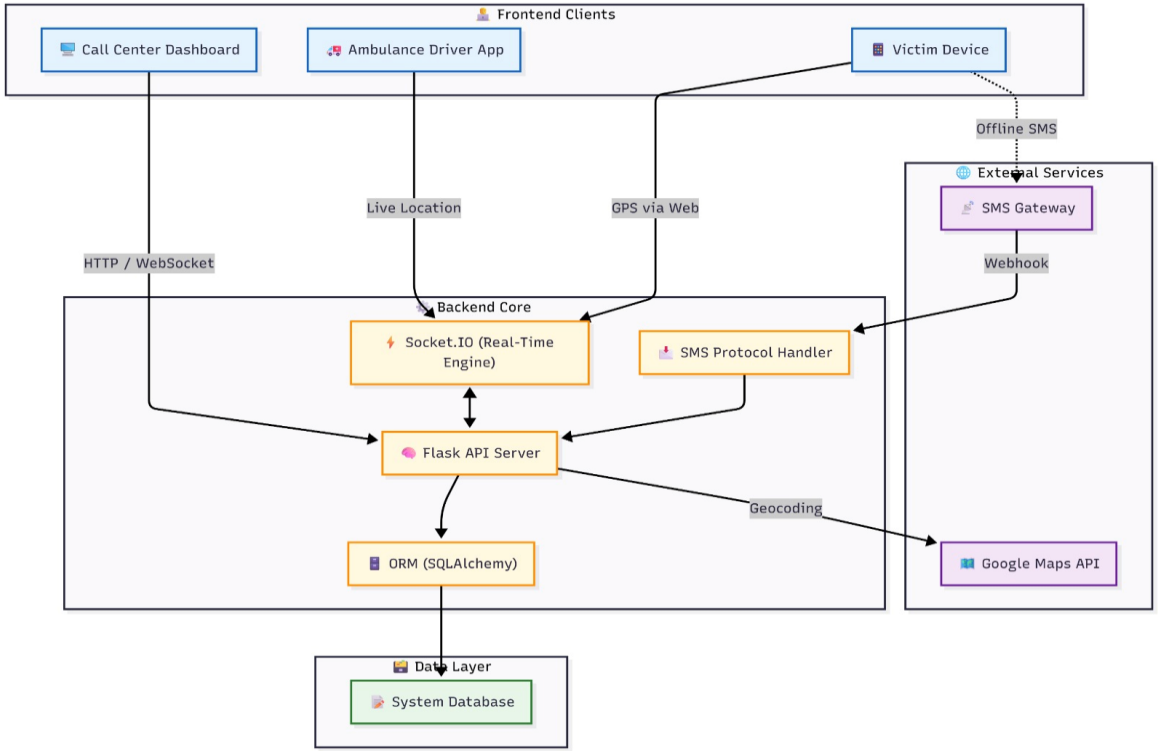
Process flow diagram or Use-case diagram



Wireframes/Mock diagrams of the proposed solution (optional)



Architecture diagram of the proposed solution



Technologies to be used in the solution

Backend & Core Services

- Python (Flask) with Socket.IO enables secure, real-time emergency coordination.

Data Management

- SQLite with SQLAlchemy ORM ensures reliable and structured system data.

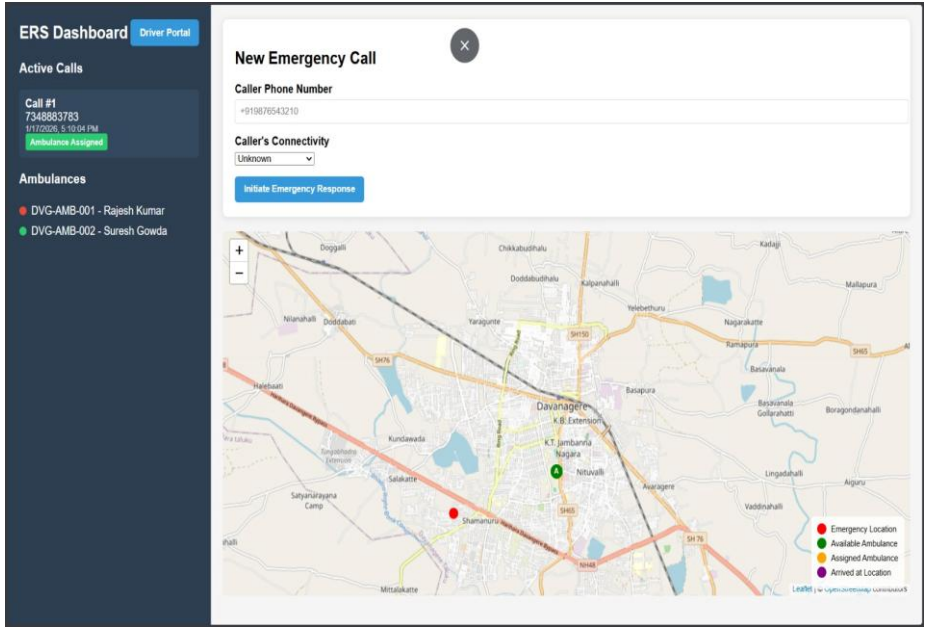
Frontend Experience

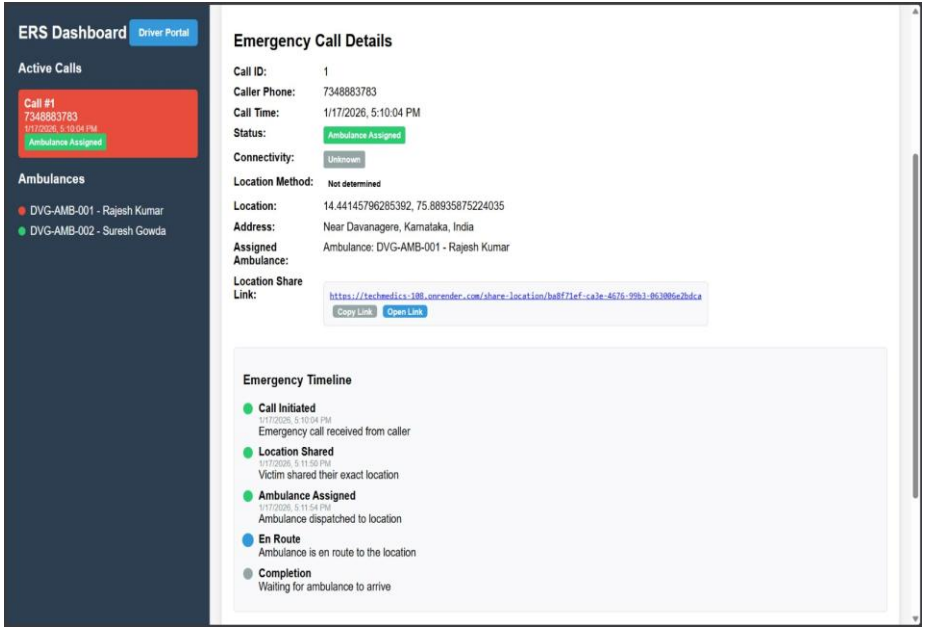
- HTML5, JavaScript, and CSS3 deliver a responsive Glassmorphism-based user interface.

Mapping & Navigation

- Leaflet.js powers live, interactive location tracking and visualization.

Snapshots of the MVP





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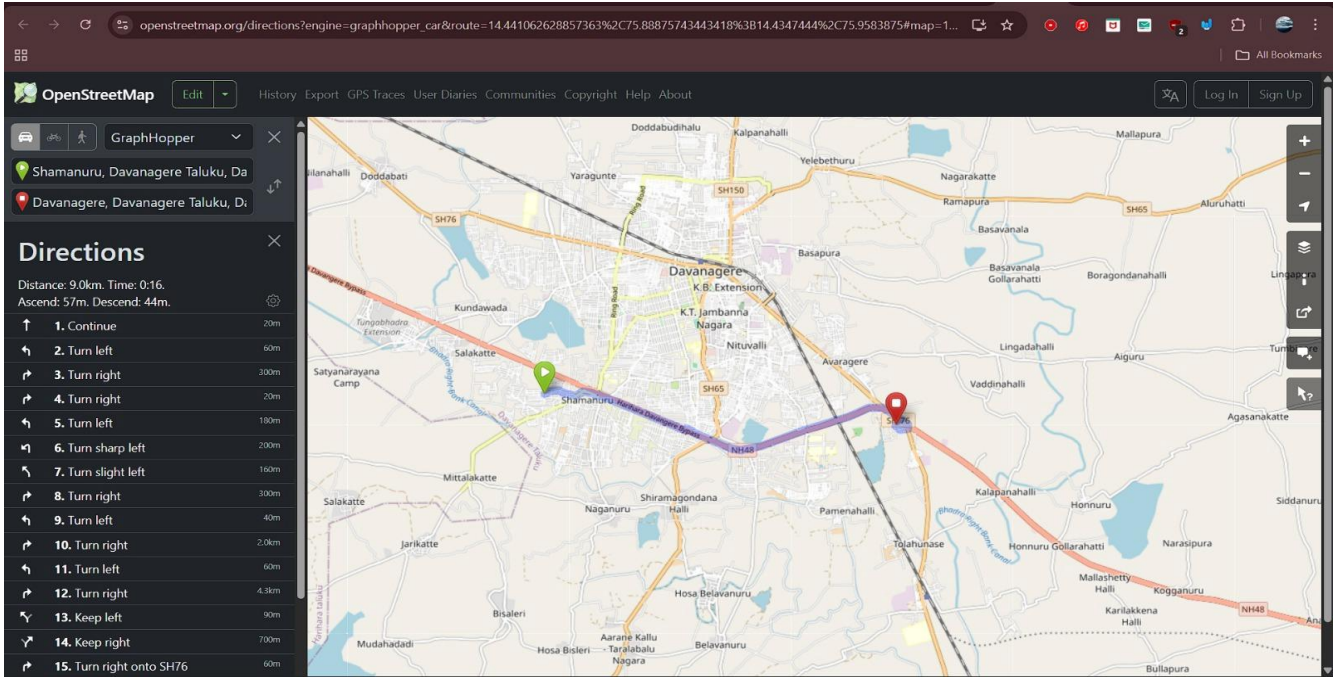

Future Development

AI Predictive Deployment
Uses historical accident data to pre-position ambulances in high-risk zones, cutting response time significantly.

Smart Traffic Integration
Enables automated green corridors by syncing with Smart City traffic signals for faster ambulance movement.

Vernacular Voice Assistance
Provides hands-free, AI-powered voice support in regional languages for rural and highway drivers.

Smart Ambulance (IoT Telemetry)
Streams real-time patient vitals to hospitals enroute, allowing emergency teams to prepare in advance.

Provide links to your:

1. GitHub Public Repository
2. Demo Video Link
3. MVP Link
4. Working Prototype Link

Build with AI

H2S HACK2SKILL

Solution Challenge

Thank You